

UEI743: BIOSENSORS AND MEMS

L	T	P	Cr
3	0	0	3.0

Course Objectives: To introduce the concept of biosensors and MEMS, design and fabrication, types and their applications. To explain biosensors and bioelectronics devices. To introduce MEMS technology.

Overview of biosensors and their electrochemistry: Molecular reorganization: Enzymes, Antibodies and DNA, Modification of bio recognition molecules for Selectivity and sensitivity, Fundamentals of surfaces and interfaces

Bioinstrumentation and bioelectronics devices: Principles of potentiometry and potentiometric biosensors, Principles of amperometry and amperometric biosensors, Optical Biosensors based on Fiber optics, FETs and Bio-MEMS, Introduction to Chemometrics, Biosensor arrays; Electronic nose and electronic tongue

MEMS Technology: Introduction Nanotechnology and MEMS, MEMS design, and fabrication technology – Lithography, Etching, MEMS material, Bulk micromachining, Surface micromachining, Microactuator, electrostatic actuation, Micro-fluidics.

MEMS types and their applications: Mechanical MEMS – Strain and pressure sensors, Accelerometers etc., Electromagnetic MEMS – Micromotors, Wireless and GPS MEMS etc.

Magnetic MEMS – all effect sensors, SQUID magnetometers, Optical MEMS – Micromachined fiber optic component, Optical sensors, Thermal MEMS – thermo-mechanical and thermo-electrical actuators, Peltier heat pumps.

Course Learning Outcomes (CLO): After the completion of the course student will be able to:

1. Exhibit the knowledge of the concept of molecular reorganization, fundamentals of surfaces and interfaces
2. Elucidate the principles of different types of biosensors.
3. Demonstrate the knowledge of the concept of MEMS design, and fabrication technology
4. Demonstrate the knowledge of bioinstrumentation and bioelectronics devices.
5. Implement different types of MEMS for biomedical applications

Text books:

1. Gardner, J.W., *Microsensors, Principles and Applications*, John Wiley and Sons (1994).
2. Kovacs, G.T.A., *Micromachined Transducer Sourcebook*, McGraw Hill (2001).
3. Turner, A.P.F., Karube, I., and Wilson G.S., *Biosensors Fundamentals and Applications*, Oxford University Press (2008).

Reference Book:

1. Trimmer, W., *Micromechanics and MEMS*, IEEE Press (1990)

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional (May include Assignments/Projects/ Tutorials/ Quizzes)	